Syllabus Details

Syllabus ID:	12039
Syllabus Name:	**Database Systems_Các hệ cơ sở dữ liệu**
Course Name English:	**Database Systems**
Subject Code:	**DBI202**
Learning-Teaching Method:	Social Constructivism,In-class lecture,Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h final exam + 85' practical exam + 104h self-study
Pre-Requisite:	
Description:	- Knowledge about database systems has become an essential part of an education in computer scien - The content of this course includes aspects of database management basic concepts, database desi
StudentTasks:	- Students must attend at least 80% of contact slots in order to be accepted to the final examination. - Student is responsible to do all exercises given by instructor in class or at home and submit on time - Promptly access to the https://flm.fpt.edu.vn/ for up-to-date course information
Tools:	Microsoft SQL Server
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	1286/QĐ-ĐHFPT dated 11/22/2024
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	11/22/2024

5 material(s)

No.	Material Description	Author	Publisher	Pub
1	First Course in Database Systems	Jeffrey D. Ullman, Jennifer Widom	Pearson	200
2	Database Management System	Raghu ramakrishnan	Mc Graw-Hill	
3	Database Management System (DBMS): A Practical Approach	Rajiv Chopra	SChand Publications	
4	Course slides (.pptx)			
5	Labs & assignment			

7 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Understand thedatabase concepts and database management system software
2	CLO2	Understand the relation model of data and Algebraic Query Language
3	CLO3	Understand data normalization and apply normalization techniques in database design
4	CLO4	Be able to model an application’s data requirements using conceptual modeling tools like ER diagrams and design
5	CLO5	Be proficient in structure query language including Data Definition Language(DDL) and Data Manipulation Langua

No.	CLO Name	CLO Details
6	CLO6	Understand PL/SQL concepts and manipulate with View, Cursors, Stored Procedures, Functions, Database Trigge
7	CLO7	Apply the Index in database design and query optimization

View mapping of CLOs to PLOs

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Session	Topic	Learning-Teac
1	Chapter 1. The Worlds of Database Systems 1.1 The Evolution of Database Systems 1.2 Overview of Database Management System	Offline
2	1.3 Outline of Database-System Studies Assignment Introduction (individual)	Offline
3	Chapter 2. The Relational Model of Data 2.1 An Overview of Data Models	Offline
4	2.2 Basics of the Relational Model	Offline
5	2.4 An Algebraic Query Language	Offline
6	2.4 An Algebraic Query Language (cont.)	Offline
7	Lab1 assistance	Offline
8	Lab1 assistance (cont.)	Offline
9	Chapter 3. Design Theory for Relational Databases 3.1 Functional Dependencies	Offline
10	3.2 Rules About Functional Dependencies	Offline
11	3.3 Design of Relational Database Schema	Offline
12	3.3 Design of Relational Database Schema (cont.)	Offline
13	3.5 Normal Forms (1NF, 2NF)	Offline
14	3.5 Normal Forms (cont. 3NF, BCNF)	Offline
15	Lab 2 assistance	Offline
16	Lab 2 assistance (cont.)	Offline
17	Progress test 1 (<=30') Assignment assistance	Offline
18	Assignment assistance (cont.)	Offline
19	Chapter 4. High-Level Database Models 4.1 The Entity / Relationship Model	Offline
20	4.2 Design Principles 4.3 Constraints in the E / R Model	Offline
21	4.4 Weak Entity Sets 4.5 From E / R Diagrams to Relational Models	Offline
22	4.6 Converting Subclass Structures to Relations	Offline
23	Lab 3 assistance	Offline
24	Lab 3 assistance (cont.)	Offline
25	Assignment assistance	Offline
26	Assignment assistance (cont.)	Offline

Session	Topic	Learning-Teac
27	Chapter 6. The Database Language SQL 6.1 Data Definition Lanaguage (DDL)	Offline
28	DDL (cont.)	Offline
29	6.2. Implement constraints on attributes with MS SQL Server (Keys and Foreign Keys, UNIQUE, CHECK,..)	Offline
30	6.2. Implement constraints on attributes with MS SQL Server (cont.)	Offline
31	6.3 DML introduction & Basic of SQL Queries	Offline
32	DML (cont.)	Offline
33	6.4 Query on more one relation	Offline
34	6.4 Query on more one relation (cont.)	Offline
35	6.5 Nested Queries in SQL	Offline
36	6.5 Nested Queries in SQL (cont.)	Offline
37	6.6 Aggregation Queries in SQL	Offline
38	6.6 Aggregation Queries in SQL (cont.)	Offline
39	6.7 Database Modifications (INSERT, UPDATE statement)	Offline
40	6.7 Database Modifications (cont. DELETE statement)	Offline
41	Chapter 7. Practical Issues of database application 7.1. Index	Offline
42	Index (cont.)	Offline
43	7.2. Transaction, View and Query Optimization	Offline
44	7.2. Transaction, View and Query Optimization (cont.)	Offline
45	Lab 4 assistance	Offline
46	Lab 4 assistance (cont.)	Offline
47	Assignment assistance (focus on query)	Offline
48	Assignment assistance (focus on query) (cont.)	Offline
49	Chapter 8. Constraints and T-SQL Programming 8.1 Triggers	Offline
50	8.2 Constraint with triggers	Offline
51	8.3 View, Function	Offline
52	8.3 View, Function (cont.)	Offline
53	8.4 Cursors	Offline
54	8.5 Implement stored procedure with MS SQL Server	Offline
55	Progress test 2 Lab 5 assistance	Offline
56	Lab 5 assistance (cont.)	Offline
57	Assignment review	Offline
58	Assignment review (cont.)	Offline
59	Review	Offline
60	Review	Offline

30 Constructive question(s)

1	1	CQ1.1	Why do we need database? How many stages of upgrade or development has the database gone throug
2	2	CQ1.2	What is a Database System and the role of Database Systems? What is a Database Management System
3	3	CQ1.3	What is Data Model? What belong to a Data Model?How can we describe and give some examples of th
4	4	CQ2.1	What are Relation instances, Schema, DB schema. How can we use SQL to define relation schemas?
5	5	CQ2.2	What is an Algebra? What is Relational Algebra? Why do we need Algebraic Query Language?
6	6	CQ2.3	What are usecases of Relational Algebra Operations?
7	7	CQ3.1	What are constraints on Relations?
8	8	CQ3.2	How can we express constrains by relational algebra?
9	9	CQ3.3	How can we use relational algebra to define constrains in some database?
10	13	CQ5.1	What are Functional Dependencies? Why you need rules about Functional Dependencies? What is useca
11	14	CQ5.2	What are keys, super-keys, closure-sets, closing sets of FD’s and how to determine them?
12	15	CQ5.3	What are Anomalies? Can we solve Anomalies? How to solve?
13	16	CQ6.1	Why should we decompose a relation?
14	17	CQ6.2	What is a simple way to normalize schema to 1NF, 2NF, 3NF, BCNF? Please, give some examples.
15	18	CQ6.3	What are the good, bad and ugly of decomposition?
16	22	CQ8.1	What is Database model? How can we describe the Database Design Process?
17	23	CQ8.2	How many Level Database Model? What are they?
18	24	CQ8.3	Why do we use Conceptual Design (or Logical Design)?
19	25	CQ9.1	What are Entity, Entity Set, Entity Instances, Attributes and others in ERD?
20	26	CQ9.2	What is a relationship and how to realize it? How can we draw an ERD?
21	27	CQ9.3	Are there any principles or constraints in designing ERDs?
22	34	CQ12.1	How can we use simple queries in some examples?
23	35	CQ12.2	How can we query data involving more than one relation? Please, give some examples.
24	36	CQ12.3	What are subqueries? Can you apply them im some examples?
25	37	CQ13.1	How can we use grouping and aggregate functions? Please, give some examples.
26	38	CQ13.2	What is HAVING clause? When should we use grouping and aggregate functions with HAVING clauses?
27	39	CQ13.3	What are DDL, DML? How can we describe INSERT, UPDATE, DELETE queries? Please, give some exam
28	46	CQ16.1	How can we use SQL to create/modify constraints of a relation?
29	47	CQ16.2	What is a trigger? Why do we use triggers?
30	48	CQ16.3	How many components in a trigger? How to create a trigger?

5 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type
1	Assignment	on-going	1	20.0%	>0	28 slots		
2	Lab	on-going	5	10.0%	>0	90'		
3	Practical Exam	on-going	1	30.0%	>0	85'		

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type
4	Progress test	on-going	2	10.0%	>0	30'		Multiple choices; Marked by Computer or a
5	Final exam	Final exam	1	30.0%	4	60		